# EXECUTION-PATH QUALIFICATION AND REALIZED COSTS IN SPECULATIVE DECODING ON CONSUMER SYSTEMS

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## ABSTRACT

Local speculative-decoding opportunity establishes neither an output-preserving path nor integrated speedup. Fresh direct-ID studies find that serialization qualifies RTX at 90/90 pairs but leaves B570 blocked at 78/90. On the tested B570 stack, a target-only micro-batch-one predictor frozen before live outcomes matches all 72/72 generated-ID sequences within a 128-ID/EOG endpoint: 24 repeat-stable reserved prompts, three runs each. Four prompts (12 runs) diverge from the reference, which matches only 60/72 live runs on the same set. This is predictive target-partition sufficiency for this path, not a unique numerical cause or a tested live-loop intervention. On 358H, an isolated NPU draft advantage does not survive the combined replay-plus-proposal interface; a later intervention stops at qualification. On 60 held-out M4 prompts, Guard and fixed speculation preserve 180/180 matched sequences. Guard reaches 1.3114× post-prefill speedup over target-only, but fixed speculation is faster (1.3469×); no incremental control benefit is shown. The cases separate output qualification, frozen trajectory prediction, and realized integration cost.

## 1 INTRODUCTION

Cheap drafting and batched target validation motivate speculative decoding (Leviathan et al., 2023). Consumer hardware already shows gains and slowdowns without verification parallelism (Chordiya, 2026). We ask whether an execution path preserves the intended output, whether a frozen predictor can reproduce a blocked path's trajectory, and whether local opportunity survives realized cost.

Two prospectively frozen direct-ID studies use shared HumanEval30 task IDs, the Qwen3-8B/0.6B Q8_0 pair, greedy $k = 2$, three repetitions and a 90/90 gate. RTX changes from 75/90 batched pairs to 90/90 serialized pairs; B570 changes from 60/90 fast to 78/90 serialized pairs and remains blocked. These fresh gates do not certify historical timing. On the tested B570 stack, target-only micro-batch one reproduces 36/36 exposed discovery sequences. Frozen before reserved live outcomes, it then predicts 72/72 capped complete generated-ID sequences: 24 repeat-stable prompts with three runs each, including four reference-divergent prompts (12 runs). The reference itself matches 60/72 on that same set. This is predictive sufficiency for the tested path, not unique numerical causality or a live-loop partition intervention.

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Preliminary work. Under review by the Machine Learning and Systems (MLSys) Conference. Do not distribute.

M4, RTX 5070 Ti Laptop, Arc B570, and 358H (Table 1) are separately frozen consumer-system case studies, not a matched hardware ranking. On 358H, the isolated NPU draft advantage does not survive the tested stateful integration, while complete target-side work dominates CPU drafting. Separate M4 representative and preregistered HumanEval campaigns yield 0.842× and 1.387× under historical text-path comparators; differing sessions preclude workload-only attribution.

Heterogeneous placement, cost prediction, numerical path dependence, deterministic inference, and adaptive speculation are established (Ruiz y Mesa et al., 2026; Wang et al., 2026; Yuan et al., 2025; Gond et al., 2026; Liu et al., 2026a). Our three contributions are (1) fresh direct-ID execution-path qualification on RTX/B570, separate from historical timing; (2) a prospectively frozen B570 predictor matching 72/72 reserved executions across 24 prompts, four of them reference-divergent, on this stack; and (3) direct characterization of realized 358H state-interface cost.

For practitioners, the bounded evaluation sequence is to qualify the intended output path, test target-partition sufficiency if blocked, then measure complete realized cost on a qualified path.

## 2 EXPERIMENTAL QUESTION AND MEASUREMENT DISCIPLINE

We investigate the gap between favorable component measurements and integrated post-prefill decode outcomes.

Acceptance describes candidate agreement, isolated draft throughput describes one execution boundary, and target-forward batching describes another. None alone measures the complete speculative loop. We use an output contract and an accounting lens to make those boundaries explicit before comparing the frozen campaigns.

### 2.1 A path-qualified output contract

A greedy speculative round proposes a draft sequence, verifies it with the target, and commits the matching prefix followed by a correction or bonus token when the remaining output budget permits. Rejected candidate state must be removed or reconstructed consistently. Our intended output contract requires generated token IDs to equal the matched target-only greedy output under the frozen configuration. This makes output preservation an explicit requirement and compares latency for the same generated sequence. It is an empirical property of a model, runtime, execution path, and workload, not a property inferred from the algorithm name.

We use three output-evidence classes throughout: historical text-path agreement (H), fresh direct-ID qualification (Q), and frozen predictor-to-live complete-ID agreement (P). H compares arrays reconstructed from parsed CLI text and cannot establish direct generated-ID equality. Q compares captured generated IDs with a frozen reference; P asks whether a frozen predictor reproduces reserved live serialized IDs. The 358H and prospective M4 studies also compare generated IDs directly. The historical counts retain their recorded comparator.

We call a path *admissible* only within its tested workload and recorded comparator. For historical text-path comparisons this denotes campaign qualification, not certification under the direct-ID contract. A failed pair blocks a workload-global optimization claim even if the timing is favorable. A small qualification probe establishes eligibility for broader testing, not universal equivalence. We retain blocked-path timings as diagnostics with their correctness status visible. This deliberate contract is stricter than an approximate-quality deployment objective; HumanEval token equality does not establish functional correctness or pass@$k$.

### 2.2 Evidence classes

**Primary** results answer a frozen workload question after its correctness gate. **Secondary** results are registered variants or explicitly bounded follow-up controls. **Mechanism-supported** claims use path controls or timed regions. **Diagnostic** values describe correctness-blocked or otherwise restricted paths. These labels describe permitted inference, not just measurement precision. In particular, the text-path-qualified RTX serialized controls are secondary evidence; they do not replace its blocked primary fast path. Historical B570 follow-ups remain diagnostic after the formal stop; the new qualification studies have a separate prospective direct-ID endpoint.

### 2.3 Complete round costs

Let $T_b$ be the matched target-only cost per output token. The realized draft time $T_d^{\text{real}}$ denotes the realized draft-subsystem time per round, including any prefix replay performed by its interface. The complete target time $T_{\text{target,round}}^{\text{real}}$ includes all measured target-side round work: candidate evaluation and any correction, bonus, or rollback work inside that region. Residual synchronization, transfer, orchestration, and bookkeeping time not included in either region is $T_x$. Let $C$ count *all* committed output tokens. With consistent timing and aggregation boundaries,

$$S = \frac{T_b C}{T_d^{\text{real}} + T_{\text{target,round}}^{\text{real}} + T_x} \tag{1}$$

$$= \frac{C}{D_{\text{real}} + R_{\text{target}} + X}, \tag{2}$$

where each denominator term is normalized by $T_b$. Break-even requires $C > D_{\text{real}} + R_{\text{target}} + X$. These identities organize measurements; they are not a new profitability theory. Cost/commit decompositions already appear in runtime-controlled studies (Kim et al., 2026).

At an aggregate level, round means and matched total-time/token accounting must use the same weights. A ratio of total times need not equal a geometric mean of prompt speedups. Likewise, accepted draft tokens per round omit correction or bonus output and cannot automatically be used as $C$. We do not infer unmeasured replay-only, transfer-only, or correction-only components by subtraction from incompatible experiments.

### 2.4 A separate target-forward proxy

For context $n$ and path $e$, define

$$V_k^{\text{micro}}(n, e) = \frac{T_{\text{target forward}}(k, n, e)}{T_{\text{target decode}}(1, n)}. \tag{3}$$

This microbenchmark asks whether a target can process a causal block near the cost of a single decode. It omits draft generation and can omit target correction/bonus and state-management work. Consequently $V_k^{\text{micro}}$ is not automatically $R_{\text{target}}$. Likewise, isolated draft throughput cannot substitute for realized stateful draft cost. These distinctions guide diagnosis after the observed outcomes; they do not supply a universal explanation for those outcomes.

*Table 1.* Independently frozen case studies; not a matched hardware comparison. All final target models are Qwen3-8B Q8_0. Greedy execution uses campaign-matched target-only references; historical M4/RTX/B570 use re-tokenized text, while 358H uses direct IDs.

| System | Runtime / target backend | Draft model / placement | Evidence scope |
|---|---|---|---|
| MacBook Air M4 | llama.cpp b10809 / Metal | Qwen3-0.6B Q8_0 / Metal | Representative 24Q and separate HumanEval30 |
| Core Ultra 9 275HX + RTX 5070 Ti Laptop | llama.cpp b10839 / CUDA | Qwen3-0.6B Q8_0 / CPU or same GPU | Blocked fast path; text-path-agreeing controls |
| Core i5-13400 + Arc B570 | llama.cpp b10839 / Vulkan0 | Qwen3-0.6B Q8_0 / B570 | Formal gate failure; later diagnostics |
| Core Ultra X7 358H + Arc B390 | llama.cpp b10839 / Vulkan0; OpenVINO draft | Qwen3-0.6B symmetric INT4 / CPU or NPU | Closed Phase 3; direct stateful decomposition |

# 3 Experimental Methodology and Platforms

## 3.1 Frozen systems and workloads

The synthesis of separately frozen consumer-system cases is retrospective: campaign-level primary outcomes were frozen independently. Later path controls and mechanism analyses retain their stated secondary, mechanism-supported, or diagnostic scope; they do not form a jointly preregistered hardware comparison.

Table 1 summarizes the completed hardware campaigns. The separately frozen runtime protocols and their distinct M4 estimator appear in Section 7. Final comparisons process one request at a time (sequence batch one). The final target is Qwen3-8B Q8_0; the native draft is Qwen3-0.6B Q8_0. The 358H heterogeneous adapter instead uses the qualified symmetric INT4 OpenVINO draft. We retain this quantization difference rather than describing the stacks as identical. Earlier M4 MLX-LM precision/correctness failures belong to qualification history and are distinct from the final llama.cpp Metal path.

The M4 representative sample freezes 24 previously unseen Spec-Bench prompts (Xia et al., 2024), with four from each of six workload groups and prior diagnostic IDs excluded. It uses three randomized paired repetitions. A separate preregistration freezes 30 HumanEval IDs (Chen et al., 2021) before prompt download or inspection; fixed $k = 2$ is the primary condition. The registered confidence-threshold conditions remain secondary. The subsequent HumanEval comparisons retain the frozen code-workload scope, three paired repetitions, greedy decoding, and a 128-token cap. They do not execute generated code.

The B570 target-partition prediction follow-up separates outcome-informed discovery from reserved validation. An exposure audit conservatively excludes any HumanEval task with a recorded B570 association; 31 of 164 tasks are excluded, leaving 133 eligible. A deterministic rule selects 24 reserved tasks with no known exposure in the audited records. On 12 exposed discovery tasks, a frozen H1/H2 decision tree selects H1 only after target-only greedy micro-batch one matches all 36 serialized sequences. The hypothesis is committed before any reserved inference. All 72 complete reserved predictions are committed before the live serialized outcomes; success requires 72/72 predictor-to-live complete-ID agreement, with no post-outcome predictor change. These are three repeat-stable runs per task, not 72 independent prompts. "No known exposure" describes the audited scope, not proof about unavailable records. The appendix gives the chronology and selection details.

Primary timing compares matched decode regions, excluding model loading and initial prompt prefill. In the native campaigns we use the frozen runtime decode counters; 358H explicitly times the complete post-prefill speculative loop. Replaying the committed prefix *inside* that loop remains charged to drafting. Thus the measurements are conditional decode costs, not whole-request latency including startup. Target-forward controls have their own narrower timing boundary. Different runtime versions, quantization, and device paths prevent a hardware-only causal comparison.

## 3.2 Statistical estimators and uncertainty

For each hardware-study prompt we reduce its three paired baseline/speculative ratios, then report the geometric mean across prompts. M4 and RTX use a within-prompt geometric mean, as does the B570 diagnostic analysis; 358H uses a within-prompt median. Intervals are 95% prompt bootstraps. M4 HumanEval uses 100,000 resamples; the other reported campaigns use 10,000. Frozen seeds and endpoint conventions are given in the separate appendix. We preserve campaign-specific estimators and finite-bootstrap input order rather than harmonizing them after observing outcomes.

These intervals quantify prompt sampling conditional on the campaign. They do not encompass all session, temperature, or runtime-state uncertainty. Randomized adjacent pairing reduces drift but does not remove it. We therefore keep the

representative and code-conditioned M4 studies separate, including their differing session conditions.

### 3.3 Provenance and energy boundaries

Frozen definitions, processed summaries, scripts, and archived evidence anchor every quantitative claim in an internal ledger. Equality counts are interpreted using the comparator distinction in Section 2, not as uniform direct-ID evidence. Historical Windows raw per-run provenance is weaker, although committed pair tables and mechanism captures support recomputation. The new RTX/B570 qualification studies retain direct-ID traces and independently replayable archives; their common protocol family and distinct settings are described in Section 5.1. The 358H reconstructed draft source is `STRONGLY_SUPPORTED`; exact historical upstream identity is not proven. The header-only blocked-preflight derivative is excluded from final analysis.

Energy is interpreted only within its qualified sensor boundary. M4 measures package aggregate excluding separate DRAM; B570 measures adapter board energy. RTX board and CPU-package domains are not a common whole-system boundary, and its serialized HumanEval follow-up lacks an admissible formal energy comparison. No end-to-end Phase-3 energy claim is made for 358H. We do not rank platform energy efficiency.

## 4 Gains Do Not Translate Uniformly

Table 2 reports historical timing under recorded text-path agreement for M4/RTX/B570. Fresh direct-ID qualification appears separately in Section 5.1 and Figure 3; it does not convert those timings into strict-ID performance results. A blocked path remains diagnostic.

### 4.1 M4: outcomes in two separate studies

The representative frozen M4 sample produces 0.842× speedup (95% CI [0.809, 0.877]), with 72/72 text-path-equal pairs and only 1/24 prompts faster. The separately preregistered HumanEval fixed-$k = 2$ study produces 1.387× [1.351, 1.422], with 90/90 text-path-equal pairs and 30/30 prompts faster. Figure 1 retains both; the positive code-conditioned result does not replace or pool with the negative representative result.

The same M4 supports both profitable and unprofitable observed regimes that pass the historical text-path comparator. Profitability is therefore not a simple hardware capability property in these studies. These outcomes do not establish general M4 profitability or isolate unified memory as a cause of the profitable result. These are workload-conditioned observed regimes, not a controlled estimate of workload causality: workloads and session conditions differ. The earlier discovery effect size is not promoted as a confirmatory

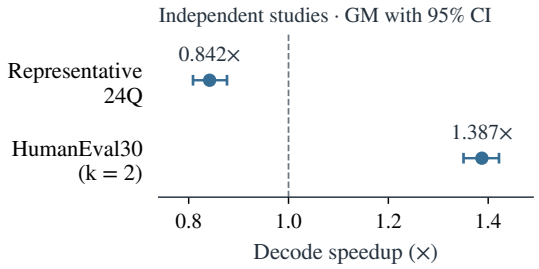



*Figure 1.* **Opposite profitability regimes on the same tested M4 stack**. Representative 24Q yields 0.842× [0.809, 0.877], while the separately preregistered HumanEval30 fixed-$k = 2$ study yields 1.387× [1.351, 1.422]. Points are campaign prompt geometric means of paired decode speedups; intervals are campaign-specific 95% prompt-bootstrap intervals. The dashed line marks parity. Both pass the recorded re-tokenized-text comparator (72/72 and 90/90 pairs); these are not direct generated-ID comparisons or functional HumanEval pass@$k$. These are separate studies, not a paired between-workload comparison; differing workloads and session conditions do not isolate workload-only causality.

estimate. Historical MLX failures also do not negate the text-path-agreeing historical Q8_0 Metal result.

For primary fixed-$k = 2$ HumanEval, the package-efficiency ratio (baseline/speculative J per output token) is 1.013× [0.990, 1.034]. The interval includes parity; a package-energy improvement is not established. The registered $p_{\min} = 0.9$ condition has secondary energy evidence, discussed separately in the appendix. The preregistered corrected synthetic ceiling was not executed. We retain that deviation rather than treating the earlier synthetic diagnostic as its replacement.

### 4.2 Across stacks: negative and blocked outcomes

On RTX 5070 Ti Laptop, the serialized CPU-to-RTX control reaches 0.404× [0.390, 0.418], while the same-GPU draft control reaches 0.650× [0.629, 0.673]. Both pass 90/90 text-path comparisons and accelerate zero of 30 prompts. The higher speed of same-GPU placement shows that placement and associated integration costs matter, but its continued loss rules out transfer as the sole explanation.

On Arc B570, neither the fast path nor the serialized fallback supports a workload-global admissible optimization result. The latter's diagnostic speedup is 0.3087× [0.2927, 0.3246], with zero of 30 prompts faster. That interval describes timing on a partially mismatching path; its precision does not turn it into an admissible speed claim. The formal failure and stop rule remain the platform result. Board-energy observations likewise remain diagnostic and boundary-limited.

On Core Ultra X7 358H, the closed Phase-3 heterogeneous

*Table 2.* Headline decode outcomes. Speedup is baseline/speculative time under each frozen prompt estimator; brackets are 95% prompt-bootstrap CIs. Above the divider, historical M4/RTX/B570 rows show recorded text-path agreement, not direct generated-ID qualification. Below it, 358H rows show direct-ID sequence equality. The historical B570 60/90 is distinct from the fresh fast-path direct-ID 60/90 in Section 5.1. Fresh RTX/B570 direct-ID gates have no matched latency and do not certify these historical timings. B570 timing remains diagnostic.

| Path / workload | Speedup [95% CI] | Endpoint | Pairs | Faster prompts | Class |
|---|---|---|---|---|---|
| M4 representative 24Q | 0.842 [0.809, 0.877] | Text-path | 72/72 | 1/24 | Primary |
| M4 HumanEval fixed-$k = 2$ | 1.387 [1.351, 1.422] | Text-path | 90/90 | 30/30 | Primary |
| CPU to RTX 5070 Ti Laptop | 0.404 [0.390, 0.418] | Text-path | 90/90 | 0/30 | Secondary |
| RTX 5070 Ti Laptop to itself | 0.650 [0.629, 0.673] | Text-path | 90/90 | 0/30 | Secondary |
| B570 serialized fallback | 0.3087 [0.2927, 0.3246] | Text-path | 60/90 | 0/30 | Diagnostic |
| 358H CPU draft | 0.734 [0.693, 0.771] | Direct-ID | 90/90 | 0/30 | Primary |
| 358H NPU draft | 0.352 [0.333, 0.369] | Direct-ID | 90/90 | 0/30 | Primary |

| Stage | Design or comparator | Complete generated-ID endpoint |
|---|---|---|
| Discovery | 12 prompts × 3 runs; select target-only UB1 as H1. | UB1 predicts the serialized trajectory in **36/36** runs. |
| Freeze | 24 reserved prompts × 3 runs; fix 72 UB1 arrays before live outcomes. | No live reserved outcome is used to select or alter H1. |
| Live test | Compare each of 72 live serialized runs with both frozen paths. | UB1 predictor/live **72/72**; UB128 reference/live **60/72 (BLOCKED)**. Four prompts (12 runs) are reference-divergent. |
| Divergent case | HumanEval/134; first difference at generated-ID index 88. | Reference ID 914 → predictor/live ID 1946 in all 3 runs. |

*Figure 2.* **B570 prospective target-partition prediction.** The 72 comparisons are repeat-stable executions of 24 distinct prompts, not 72 independent tasks. The same-set reference contrast and divergent example distinguish prediction of the observed blocked trajectory from restored qualification. H1's freeze and live ordering are reported in the appendix; complete arrays appear in the separate anonymous sequence supplement. This one-stack observational predictor is not a history-controlled live-loop intervention.

runner has direct-ID sequence equality on both CPU and NPU drafting: 90/90 pairs per path, with zero of 30 prompts faster. CPU achieves 0.734× [0.693, 0.771]; NPU achieves 0.352× [0.333, 0.369]. These direct-ID-qualified negative outcomes coexist with favorable isolated draft measurements. Section 5 investigates which measured boundaries account for the failed local promise.

## 5 From Local Opportunity to Realized Gain

The observed outcomes trace two tested boundaries: execution-path qualification and stateful realization. Figure 2 shows the B570 discovery, freeze, and reserved prediction within that sequence; the platform cases are separate studies.

### 5.1 Execution-path qualification

**Historical physical opportunity.** Figure 3A preserves eleven historical target-forward proxy cells from different runs and contexts. The M4 Metal micro ratios are $V_2^{\text{micro}} \approx 1.02$–$1.08$; RTX batching is near one baseline step while serialization costs more; B570 fast cells span 1.24–1.64 and serialized cells approximately 1.983–1.992. These proxies omit drafting and integration, cannot replace complete $R_{\text{target}}$, and do not explain the separate-session M4 contrast.

**Prospective direct-ID qualification.** Two frozen studies share 30 HumanEval tasks, three repetitions, greedy $k = 2$, and the Qwen3-8B/0.6B Q8_0 pair. Each completes 270 executions and 180 comparisons. Fresh direct-ID qualification requires 90/90 complete direct-ID sequence equality with no dropped mismatch. Different runtime/front-end, context, and reference settings preclude controlled hardware comparison or fresh performance claims.

RTX direct-ID sequence equality is 75/90 batched and 90/90 serialized, qualifying only the latter. B570 direct-ID sequence equality is 60/90 fast and 78/90 serialized; both remain blocked. The historical B570 60/90 text-path count and fresh fast-path 60/90 direct-ID count are different campaigns and comparators. All conditions are repetition-stable on 30 prompts, which does not imply reference equality. Figure 3B has no matched fresh latency endpoint and cannot

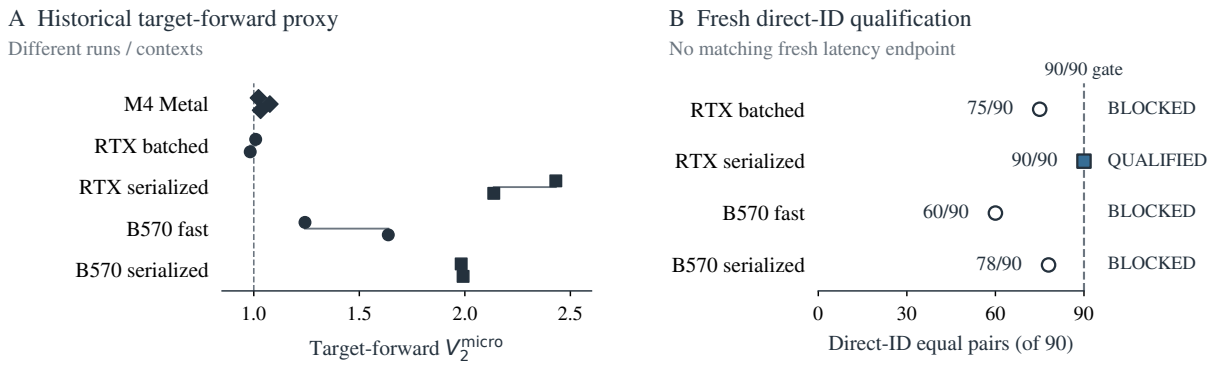



*Figure 3.* **Historical target-forward proxy and fresh direct-ID qualification are separate evidence.** A: historical $V_2^{\mathrm{micro}}$ cells from different runs and contexts; shapes encode M4 reference (diamond), fast/batched (circle), and serialized (square) paths, with no qualification encoding. Segments span measured contexts, not uncertainty. The proxy is not complete $R_{\mathrm{target}}$ or end-to-end speedup. B: fresh HumanEval30 direct-ID sequence equality, with no matching fresh latency endpoint; only RTX serialization reaches the frozen 90/90 gate. Open circles mean BLOCKED; the filled square means QUALIFIED. The panels are not a matched cost–qualification experiment. Fresh direct-ID qualification does not retrospectively authorize historical timing. Results are path/runtime scoped; contexts and references differ.

be paired with Panel A's historical cost.

**Serialization changes qualification but is not a certificate.** The prospective direct-ID results do not replace historical B570 text-path counts or certify historical RTX timing; the older RTX controls remain secondary and B570 timing diagnostic. The earlier passing small B570 probe was limited to its subset.

### 5.2 Controlled B570 target-partition sufficiency

The v0.7 B570 residual could reflect reference, front-end, context, or target-partition differences. A target-only harness first reproduces 90/90 reference sequences. On 12 outcome-exposed discovery prompts, target-only greedy execution with $n_{\mathrm{ubatch}} = 1$ matches 36/36 serialized speculative sequences. The frozen H1/H2 decision tree selects H1; H2 fixed-proposal replay is not run because H1 passes. Discovery is exploratory. H1 could still have failed to predict any complete reserved live sequence, especially the reference-divergent cases.

An exposure audit conservatively excludes 31 tasks, then deterministically selects 24 reserved HumanEval prompts from 133 eligible tasks. H1 is committed before reserved inference; all 72 complete target-only predictions are committed before any live serialized outcome. Predictor-to-live complete-ID agreement is 72/72 over three repeat-stable runs per prompt. Four prompts (12 runs) are reference-divergent, while 20 are reference-equal; predictor/live agrees on all 24 classifications and all 12 divergent first indices. The same-set UB128 reference matches only 60/72 and remains *BLOCKED*. No parameter or prediction changes after their respective freezes; no selective reruns occur. The appendix gives task IDs, divergence tokens, and audit scope.

Target-only micro-batch one thus predicts the tested serialized trajectory without live draft computation. This is observational path-level sufficiency on this B570 configuration and 128-ID/EOG endpoint. It does not isolate a unique numerical operation or show what happens when only the target partition is switched inside the live speculative loop at identical committed history. It establishes no universal UB1 rule or RTX mechanism. The 60/72 reserved endpoint is distinct from the earlier 78/90 direct-ID study. The anonymous sequence supplement permits independent equality checks on the archived arrays but does not independently authenticate the reported pre-live freeze time.

**Separate RTX numerical diagnostic.** A frozen CUDA reproducer captures repeat-stable HumanEval/106 argmax divergence, with a same-commit diagnostic score-margin reversal (appendix). This historical batch-shape observation (Yuan et al., 2025; Gond et al., 2026) does not identify a kernel, replace formal measurements, or explain B570.

### 5.3 Stateful realization

On Core Ultra X7 358H, isolated NPU draft throughput exceeds CPU at tested contexts, yet both qualified integrated paths are unprofitable and NPU is slower. Direct round records show complete mean committed progress $C = 2.4631$ and $2.4584$, respectively, rather than the older

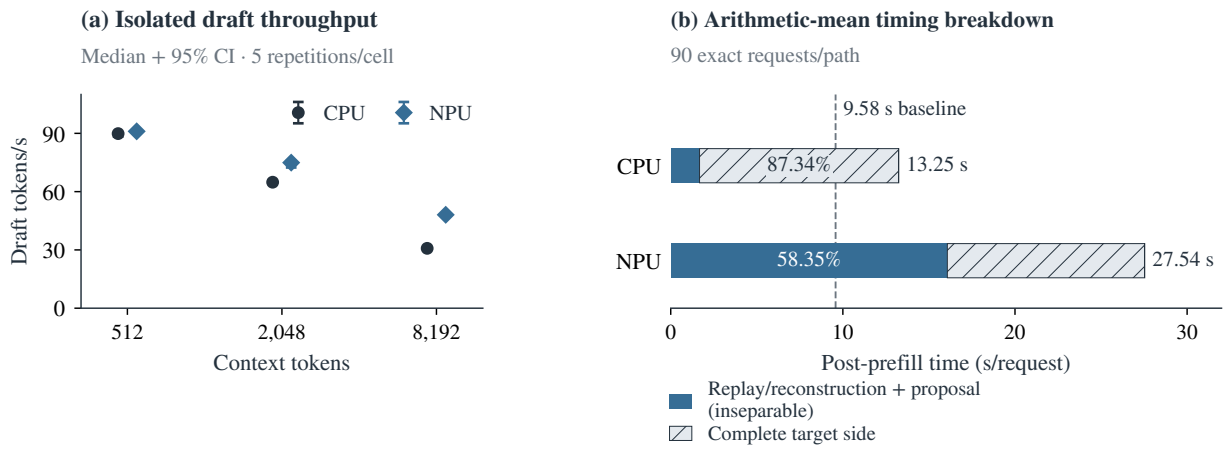



*Figure 4.* **Isolated draft advantage does not survive the tested stateful integration**. The NPU's 58.35% region is a single *full-prefix reconstruction/replay plus proposal* call; its components are inseparable and replay-only savings are unmeasured. (A) Phase-1 median steady-state draft throughput and 95% repetition-bootstrap intervals, five repetitions per cell. (B) Phase-3 arithmetic mean post-prefill seconds per request over 90 exact observations per path. CPU and NPU totals are 13.25 s and 27.54 s against a 9.58 s mean baseline. The CPU path is target-side dominated (87.34%). The 0.734× and 0.352× headlines use a separate prompt-GM estimator, not ratios of these arithmetic-mean bars. Initial prompt prefill is excluded. This diagnoses the tested interface, not intrinsic NPU capability or a measured state-persistence benefit. CPU replay plus proposal is 12.63%; NPU complete target work is 41.63%. Residuals are 4.81 ms (0.04%) and 5.06 ms (0.02%). Prompt-GM 95% CIs are [0.693, 0.771] and [0.333, 0.369]. The dashed line is the shared arithmetic-mean baseline; device marks are horizontally offset.

accepted-draft counter.

Complete target work, including persistent state, verification, rollback, and correction/bonus decoding, accounts for 87.34% of CPU-path time. On NPU, the synchronous OpenVINO call combines full committed-prefix replay/reconstruction and proposal, accounting for 58.35%; these components are inseparable in the recorded timing. Residual orchestration is 0.04% and 0.02%, respectively.

Figure 4 separates isolated throughput from realized cost. The result concerns this qualified interface, not NPU compute capability; Phase-3 timing does not estimate a persistence benefit.

Arithmetic totals reconstruct Equation 2: CPU $(D_{\text{real}}, R_{\text{target}}, X) = (0.4304, 2.9767, 0.00124)$ yields $0.7227\times$; NPU $(4.1251, 2.9435, 0.00130)$ yields $0.3477\times$. These aggregate ratios differ from Table 2's prompt-level geometric means.

The Phase-2 eligibility screen combined isolated drafting and a target-forward proxy; Phase 3 measures complete target and draft-interface cost instead. There is no Phase-3 end-to-end energy claim; source reconstruction remains `STRONGLY_SUPPORTED`.

## 6 From Measured Constraints to Runtime Decisions

Two measured boundaries constrain usable speculation. A cheap verification path can fail the output contract, and a fast isolated draft device can incur additional work through its state interface. On the tested B570 stack, the frozen target-only UB1 path was sufficient to reproduce the observed serialized trajectory without live drafting. Complete target-side work can dominate even after draft cost is lowered, as on 358H CPU drafting.

RTX serialization crosses the 90/90 gate, while B570 serialization improves agreement without crossing it. First qualify the intended path against its output contract. If it fails, test the target execution partition before assigning the failure to live drafting or device placement. Only then use complete realized cost to assess profitability on a qualified path. This evidence-derived sequence is bounded by the tested configurations. The separately frozen runtime studies on different stacks are not an intervention/ablation chain.

## 7 Qualification and Realized-Cost Runtime Control

We examine qualification and continuation in separately frozen runtime studies. The 358H candidate stops at direct-

ID qualification. On M4, a prospectively direct-ID-qualified path permits a test of a simple cost rule: Guard remains faster than target-only, but fixed speculation is faster in the post-study paired secondary comparison; no incremental control benefit is shown. These are complementary tests of the decisions in Section 6, not two successful optimization interventions.

### 7.1 An intervention stopped by qualification

The 358H component measurements motivated persistent draft state and a wider target execution setting. The implementation ran the target and draft backends in separate processes. Synchronous draft calls, including communication and state work, remained inside the target worker's post-prefill timer. The executable implementation still had to pass correctness qualification. CPU persistent-state restore/reconcile correctness could not be established under the tested representation; the Optimum NPU state-capable path was unavailable with the frozen dynamic-shape IR. These are implementation boundaries, not evidence that the devices cannot retain state.

The target setting `n_ubatch=2` (UB2) passed the small gate but failed full qualification. Correctness-only follow-up controls on Qwen3-8B Q8_0, llama.cpp b10839, and Vulkan0 reproduced different greedy tokens under UB1 and UB2 even with identical exact prefix tokens. Request-order and separate-process controls reproduced the target-only divergence. Replaying frozen proposals also reproduced the speculative divergence without live NPU draft inference. The latter depended on earlier target execution history; the controls found no position drift or specific rollback-boundary bug. The responsible floating-point or backend operation remains unidentified.

Under this frozen tested target stack, UB2 did not preserve the strict direct-ID output contract. The formal registry retained only the target reference, so the runtime study had no qualified intervention pair to evaluate. No formal R2 performance run occurred, and the follow-up diagnostics are invalid for performance claims. Qualification excluded an intervention motivated by the measured bottlenecks before its speed was evaluated.

### 7.2 A prospective cost guard on M4

The M4 experiment asks whether a frozen rule based on already-realized complete cost can control continuation while preserving exact output and profitable execution against target-only. It does not assume that adding control improves on fixed speculation. We retain the Qwen3-8B/0.6B Q8_0 pair on llama.cpp Metal, fixed $k = 2$, greedy generation, and a 128-token cap. Guard begins on the same speculative path as Always-Spec and can fall back permanently to target-only within a request. It changes neither proposal width nor device placement.

At each eligible completed round, Guard divides all elapsed charged generation cost by committed output progress $C$. This count includes the initial target seed and committed correction or bonus tokens, but excludes rejected proposals. Guard continues if this cumulative cost per token is at most $(1 - m)R_{\mathrm{ref}}$, a target-only reference frozen from calibration; otherwise it falls back. If a decision is reached with zero committed progress, Guard also falls back. Before each control decision, invalid candidate suffix state is removed from the target and draft contexts, preserving committed history for target-only continuation. The enclosing post-prefill clock includes generation-loop initialization, seed-token sampling, drafting, target work, synchronization, state reconciliation, control, and transition costs. Model loading, sampler/speculator setup, initial prompt prefill, and the speculative begin hook precede the timer; the result is not whole-request latency.

The reference is the median across calibration prompts of each prompt's median target latency divided by its median output-token count. The preregistered candidate grid varies the first decision round and margin. Selection minimizes calibration geometric-mean latency relative to the offline minimum of target-only and Always-Spec among exact candidates, with frozen tie-breaks. The selected rule uses margin 0 and first decides after 8 completed speculative rounds; it then decides after each later round while speculation continues. A request ending earlier makes no decision. The online policy receives neither task/family labels nor future output length, current-request target-only latency, or oracle outcomes.

### 7.3 Held-out evaluation

Historical M4 outcomes informed the study design. Prospective here means that the policy was frozen before the reserved prompt outcomes were measured, not that the design was uninformed by earlier campaigns. IDs marked as previously executed or outcome-exposed in the frozen inventory were excluded before deterministic hash-ranked selection. Disjoint splits contain 40 calibration prompts, 16 qualification prompts, and 60 held-out prompts, with equal HumanEval and Spec-Bench counts within each split. The policy and reference were frozen before qualification and formal measurement; the appendix documents the audited recovery of a freeze-hash metadata defect. Both Always-Spec and Guard passed 48/48 direct-token comparisons in qualification.

The prospective M4 study runs 3 repetitions of target-only, Always-Spec, and Guard on each held-out prompt, randomizing condition order within prompt and repetition: 540 observations. Both speculative conditions preserve 180/180 complete direct-ID sequences, with zero mismatches. For

*Table 3.* M4 per-prompt median post-prefill latencies. Each speculative condition has 180/180 direct-ID sequence equality. The first two rows report speedups. The offline oracle uses the lower target-only or Always-Spec median and is unavailable online; its row reports latency regret. The separated Guard/Always-Spec contrast is reviewer-requested post-study paired secondary analysis; values above one mean Guard is slower.

| Latency ratio | GM | 95% prompt CI |
|---|---|---|
| Target / Guard | 1.3114 | [1.2487, 1.3787] |
| Target / Always-Spec | 1.3469 | [1.2867, 1.4088] |
| Guard / offline oracle | 1.0278 | [1.0137, 1.0446] |
| *Post-study paired secondary latency ratio* | | |
| Guard / Always-Spec | 1.0271 | [1.0130, 1.0436] |

each prompt and condition, analysis takes the median latency across repetitions, then the geometric mean of prompt latency ratios. This ratio-of-medians estimator differs from the historical M4 text-path hardware estimator. Confidence intervals use 10,000 prompt bootstrap resamples with seed 20260924; repetitions are not independent sampling units.

Guard is profitable against target-only on this held-out workload (Table 3). Fixed speculation has the higher aggregate speedup point estimate. A reviewer-requested post-study paired secondary analysis of the frozen prompt medians gives Guard/Always-Spec latency 1.0271 [1.0130, 1.0436]. The interval lies above parity: Guard had higher latency on this comparison. By prompt median, Guard is faster on 24 prompts, Always-Spec on 36, with 0 ties. Guard falls back in 30.6% of its 180 request executions; this measures control activity, not the fraction of beneficial decisions. Guard's geometric-mean latency relative to the offline oracle is 1.0278×. That oracle chooses the lower target-only or Always-Spec median after measurement and was unavailable to the runtime. It is a descriptive hindsight comparator, not an achievable-policy guarantee.

Guard preserves direct-ID output and is profitable against target-only, but the post-study paired comparison gives no incremental benefit over Always-Spec. Neither fallback activity nor the oracle ratio establishes risk reduction or threshold optimality. The experiment supplies no evidence of policy transfer to another stack. Small individual SpecBench strata do not support stable subgroup estimates. No runtime energy claim is made. The separate appendix documents the archived recomputation and audited freeze-hash metadata exception.

## 8 Related Work

**Performance and heterogeneous execution.** Speculative decoding amortizes target computation under its algorithmic assumptions (Leviathan et al., 2023). Prior work already documents verification and workload limits: Liu et al. (2026b) study verification dominance and batch effects in vLLM; Kim et al. (2026) account for conditional execution cost and runtime control; Lyu et al. (2026) derive edge–cloud profitability conditions. On consumer Apple silicon, Chordiya (2026) report gains and slowdowns across five backend configurations, diagnose ineffective batch-parallel verification in one quantized Metal path, test output agreement, and observe target-only batch-shape argmax changes. Slowdown and correctness gating are therefore established background. Heterogeneous placement is also established: Dovetail, DuoDecoding, and Mirror-SD study GPU/CPU or GPU/NPU paths (Zhang et al., 2025; Lv et al., 2025; Bhendawade et al., 2025); Ruiz y Mesa et al. (2026) combine an analytical cost model with compiler-oriented CPU/GPU partitioning, and Lever jointly optimizes draft, verification, and CPU/NPU execution for a smartphone target (Wang et al., 2026). State reuse has prior art in compressed draft KV memory (Yuan et al., 2026). Our 358H evidence instead accounts for one measured, combined replay-plus-proposal interface; it does not claim a general placement or persistence result.

**Numerical paths and output contracts.** Zhang et al. (2026) distinguish ragged-batch synchronization errors from residual numerical mismatches. Yuan et al. (2025) study output variation across system configurations, and LLM-42 addresses deterministic inference through verified speculation (Gond et al., 2026). Numerical path dependence is not new here. Our distinct test prospectively qualifies specific RTX/B570 paths against captured generated IDs, separately from historical text-path timing. For B570, H1 was selected after 36/36 exposed discovery matches, then complete predictions were frozen before 72 reserved live executions; four prompts could have contradicted a reference-echo explanation and instead matched the non-reference prediction in all 12 runs. H2 was unrun after H1 passed. This tests path admissibility and predictive sufficiency, not a new numerical phenomenon, unique kernel cause, or competing reserved predictor.

**Adaptive speculation and the study boundary.** TurboSpec adjusts speculation for serving goodput (Liu et al., 2025); BanditSpec selects configurations online (Hou et al., 2025); ACL HeteroSpec adapts verification from contextual entropy (Liu et al., 2026a). Adaptive control is established prior work. Our M4 Guard is a qualified runtime control test, not a new policy advantage: it beats target-only, but fixed speculation is faster in the post-study paired secondary comparison. The paper's central evidence is prospective direct-ID qualification, frozen held-out complete-trajectory prediction on B570, and realized state-interface accounting on 358H.

## 9 LIMITATIONS

The four consumer systems use a narrow Qwen3 model family, with hardware, operating systems, runtime versions, backends, and some quantization choices co-varying. The cases establish specific outcomes and within-platform execution-path evidence, not an architectural ranking, a population failure rate, or controlled portability. The two diagnoses do not causally explain the separate-session M4 contrast. Serving concurrency, production load, and broader model families are outside the measured scope.

The strict output contract is conditional on the frozen reference; another quality objective could admit other paths. Historical M4, RTX, and B570 text-path agreement counts use re-tokenized CLI text, with campaign-specific parsing. They cannot establish direct generated-ID equality; the historical labels and counts remain evidence of their recorded comparator only. Two fresh RTX/B570 studies supply direct generated-ID evidence without retroactively certifying historical timing. They cover one model pair, HumanEval30, fixed $k = 2$, and three repetitions; differing runtime/reference/context settings prevent estimating a causal hardware difference. The fresh RTX driver is 616.92 versus historical 616.56. B570 launcher recovery changes deployment, not frozen study semantics. The H1 predictor covers one B570 platform, model pair, runtime/build, greedy $k = 2$, HumanEval workload, and 24 repeat-stable reserved prompts, only four of which diverge from the reference. "No known B570 exposure" means no record in the audited scope, not proof that a task was never run. Target-only UB1 predicts these trajectories but does not establish a history-controlled live-loop partition effect, a specific kernel, an architecture-wide cause, a CUDA/B570 common mechanism, or a universal micro-batch rule. The anonymous sequence supplement permits recomputing equality but not independently authenticating the recorded freeze time; the PDF alone displays digests and reported chronology, not all full arrays. Prompt counts do not establish backend-wide behavior. HumanEval functional execution and pass@$k$ were not evaluated. CUDA/B570 and 358H findings concern their tested paths, not general backend unsuitability. Replay and proposal share one timed region, so individual causal costs and a persistence benefit are not established. The 358H intervention did not reach formal performance evaluation.

Campaign estimators and bootstrap conventions differ. Prompt intervals do not cover all session/thermal uncertainty. Timing excludes initial prefill and loading and is not whole-request user latency. Energy domains differ, the corrected M4 synthetic-ceiling preregistration remains unexecuted, and no common energy ranking or end-to-end 358H Phase-3 energy claim is supported.

The runtime result covers one M4 model/runtime pair, 60 held-out prompts, fixed $k = 2$, and one-way fallback. The paired Guard/Always-Spec contrast is post-study secondary analysis, not the registered primary endpoint; its prompt-bootstrap uncertainty excludes session and recalibration variation. Guard had higher latency on this frozen comparison. Policy transfer, stable small-stratum conclusions, and runtime energy benefit remain unestablished. Historical outcomes informed the design; policy freezing protects the reserved prompt evaluation, not ignorance of earlier platform results.

Historical Windows raw per-run provenance is weaker than in the other campaigns; the new direct-ID studies retain independently validated raw evidence. The 358H draft-source reconstruction is `STRONGLY_SUPPORTED`, not proven historical identity, and its runtime diagnostic is available as a compact record without local raw-trace replay. The M4 freeze-hash metadata defect was independently reconciled against the actual freeze, formal-start commit, and archived observations. The erroneous original markers and pipeline stop remain intact; the appendix explains this recovery rather than treating it as uninterrupted pipeline validation.

## 10 CONCLUSION

Fresh direct-ID studies qualify RTX serialization but leave B570 blocked, without a matched fresh latency endpoint. On the tested B570 stack, a frozen target-only UB1 predictor matches 72/72 capped generated-ID runs across 24 repeat-stable prompts; four prompts (12 runs) diverge from the reference. This is predictive sufficiency on that path, not a unique cause or controlled live-loop intervention. On 358H, the combined replay/reconstruction-plus-proposal interface consumes an isolated draft-device advantage; its later intervention stops at qualification. M4 Guard preserves direct-ID output and beats target-only (1.3114×), but fixed speculation (1.3469×) wins the post-study paired comparison; no incremental control benefit is shown.

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